ODR Thesis: The Compton Count as the Only Primitive — A Five-Question Synthesis

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2026-08-03

Abstract

The Ontological Distribution of Reality (ODR) framework identifies a particle with its Compton count $N_C = m/m_P \in \mathbb{Q}^+$, a dimensionless rational number whose prime factorization assigns it a fixed vertex on the Bruhat–Tits tree. This paper synthesises five research questions into a single thesis: (RQ1) every physical quantity — mass, length, time, action, and all Standard Model coupling constants — can be expressed as a rational function of N_C [speculative — requires extension to interacting QFT]; (RQ2) the physical number system is $\mathbb{Q}$ (computable rationals), not the full Archimedean continuum $\mathbb{R}$ [speculative — driven by Ostrowski's theorem and place-democracy]; (RQ3) the Bruhat–Tits tree — a purely combinatorial, origin-less, scale-less coordinate system — can replace Cartesian $\mathbb{R}^3$ as the natural geometry of physics [speculative — most fragile claim]; (RQ4) this framework generates testable Planck-scale predictions distinguishing it from standard QFT, including Lorentz-violation dispersion relations and Compton-count quantisation in ultra-high-energy cosmic rays [speculative — five candidates prioritised]; and (RQ5) five independent disciplines (number theory, quantum foundations, information theory, mathematical physics, holography) each discovered the same $(p+1)$-valent combinatorial tree with cross-ratios between 1916 and 2016, called it by five different names, and spent 66-110 years never connecting them — a century-long silo failure that the ODR thesis rectifies [established — historical].

Keywords: Ontological Distribution of Reality, Bruhat–Tits tree, Compton frequency, p-adic valuation, place-democracy, quasiparticles, particle ontology, silo failure, consilience

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1. Introduction

What is a particle? The question is ancient but the answer remains contested. Quantum field theory (QFT) defines a particle as an excitation of an underlying field [established — standard QFT]. Condensed matter physics treats them as emergent

collective excitations — quasiparticles — whose masses depend on the host material and are not intrinsic [established — Landau 1957, Kittel 2005]. Ontologically, there is no consensus: a particle is what you can count, or what carries conserved quantum numbers, or what persists across scattering events.

The Ontological Distribution of Reality (ODR) framework proposes a radical alternative: a particle IS its Compton count $N_C = m/m_P \in \mathbb{Q}^+$, where m_P is the Planck mass [speculative — ODR v3.0, DOI 10.5281/zenodo.21755425]. The ratio N_C is a dimensionless rational number. Its prime factorisation determines a specific vertex on the Bruhat–Tits (BT) tree — a $(p+1)$-valent combinatorial graph with ultrametric distance, no absolute origin, and no inherent scale [established — mathematics; Kaletha & Prasad 2023]. Every other physical quantity — energy, wavelength, cross-sections, coupling constants — is a rational function of the Compton count.

This paper synthesises five research questions into a single thesis statement, proceeding through the Compton ontology (RQ1, §2), the number system (RQ2, §3), the coordinate system (RQ3, §4), testable predictions (RQ4, §5), and the meta-scientific silo failure diagnosis (RQ5, §6). Throughout, certainty calibration follows the qnfo-core convention: [established] for textbook facts and historically verified claims; [speculative] for claims without external precedent; [my conjecture] for the author's strongest interpretative leaps.

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2. RQ1: The Compton Count as the Only Primitive

2.1 Definition

In quantum field theory, every free particle carries a Compton frequency:

$$\omega_C = \frac{mc^2}{\hbar}$$

In natural units ($\hbar = c = 1$), this simplifies to $\omega_C = m$. The Compton frequency is the particle's "clock rate" — the frequency at which its quantum phase evolves [established — standard QFT]. For the electron, the cyclic frequency is approximately 1.236×10^{20} Hz.

ODR elevates this frequency from a property a particle happens to have to what a particle IS: the Compton count

$$N_C = \frac{m}{m_P} \in \mathbb{Q}^+$$

is the particle's identity. [speculative — ODR v3.0 §2]

2.2 Prime Factorisation and Tree Position

Because N_C is rational, it admits a prime factorisation:

$$N_C = \prod_{p \in \mathbb{P}} p^{a_p}, \quad a_p \in \mathbb{Z}$$

The set of exponents $\{a_p\}$ for all primes determines the particle's p-adic valuation $\mathrm{ord}_p(N_C) = a_p$ at every prime. These valuations assign the particle a specific depth and branching path on the Bruhat–Tits tree: a $(p+1)$-valent tree where each vertex is identified with a rational number and edges represent transitions between numbers with different p-adic valuations [established — mathematics; Kaletha & Prasad 2023].

The electron's Compton count is $N_C^{\text{electron}} \approx 4.185 \times 10^{-23}$. Its prime factorisation is finite and exact — the decimal is merely the Archimedean projection of a rational number. [speculative]

2.3 The Quasiparticle Extension

A natural objection arises from condensed matter: what of quasiparticles — phonons, magnons, dressed electrons — whose effective mass m^* depends on the host crystal? In GaAs, the electron effective mass is $m^* \approx 0.067 \times m_e$; in silicon, $m^* \approx 0.19 \times m_e$ [established — Kittel 2005]. Their Compton counts are not invariant.

The quasiparticle extension paper [QNFO, 2026, DOI 10.5281/zenodo.21768757] resolves this: a quasiparticle is not a fixed vertex on the BT tree but a rational function of the background Compton counts:

$$N_C^*(\alpha) = f(N_C^{\text{bare}}, N_C^{\text{lattice ions}}, N_C^{\text{phonon modes}}, \ldots)$$

It lacks the "ontological permanence" of a bare particle because its prime factorisation shifts with context. This is not a flaw — it is the precise geometric account of why condensed matter physicists distinguish elementary from emergent excitations. [speculative]

Falsifiability: RQ1 would be disconfirmed if a quasiparticle were found with an effective mass identical across multiple chemically distinct host lattices — a single counterexample would demonstrate context-independence in a rational function. Such an observation has not been reported in approximately 90 years of condensed matter physics. [speculative; null test held for ~90yr]

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3. RQ2: $\mathbb{Q}$ as the Physical Number System

3.1 Ostrowski's Theorem

Ostrowski's theorem (1916) proves that the only non-trivial absolute values on $\mathbb{Q}$ are the Archimedean absolute value (the usual real norm) and the p-adic absolute values for each prime p [established — mathematics]. There are no others. The real numbers $\mathbb{R}$ are the Archimedean completion of $\mathbb{Q}$; the p-adic numbers $\mathbb{Q}_p$ are the non-Archimedean completions.

Physicists have used $\mathbb{R}$ for approximately 400 years, implicitly treating the Archimedean completion as the "real" number system. ODR asks: what if $\mathbb{Q}$ itself — the rational numbers — is the primitive, and $\mathbb{R}$ and $\mathbb{Q}_p$ are merely completions? [speculative]

3.2 Place-Democracy

ODR's central ontological move is "place-democracy": no completion of $\mathbb{Q}$ is privileged. A Compton count $N_C \in \mathbb{Q}^+$ is a pure rational number — it has an Archimedean value (the mass in Planck units), AND it has p-adic valuations at every prime. Both are equally real, or rather, both are equally valid completions of the same primitive rational. [speculative]

The Continuum Trilogy Paper I [QNFO, 2026, DOI 10.5281/zenodo.21672990] establishes that the physical continuum is NOT the full real line (including non-computable reals) but the computable Archimedean continuum crossed with computable p-adic continua:

$$\mathbb{R}_c \times \prod_{p \in S} \mathbb{Q}_p^c$$

The breadth of $\mathbb{R}$ — the power-set overhang of non-computable reals — is physically unfalsifiable: no finite measurement protocol can discriminate two non-computable numbers. [established — Continuum Trilogy I] The depth — limits, continuity, dynamics — is preserved. [speculative]

3.3 The Irrational Constants Objection

The standard objection: "But what about π, e, $\sqrt{2}$?" These numbers appear in physics and are not rational. ODR's response: (a) π and e are computable reals — limits of computable rational sequences; they exist in $\mathbb{R}_c$ which is the computable subfield of the Archimedean completion; (b) in Planck units, $\pi = C/d$ (circumference-to-diameter ratio) — the decimal $3.14159\ldots$ is the Archimedean projection of a ratio, not a magical transcendental number; (c) $\sqrt{2}$ appears when solving $x^2 = 2$ — the equation itself is a rational relationship; the solution lives in the quadratic extension $\mathbb{Q}(\sqrt{2})$, which is a finite algebraic extension of $\mathbb{Q}$, not an invocation of the full continuum $\mathbb{R}$. [speculative]

Falsifiability: RQ2 would be disconfirmed if a physical quantity were found that demonstrably requires a non-computable real number for its specification and where the difference between two non-computable candidates is experimentally measurable. This has never been observed. [speculative]

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4. RQ3: The Bruhat–Tits Tree as Coordinate System

4.1 The Tree

The Bruhat–Tits tree $\mathcal{T}_p$ for a prime p is a $(p+1)$-valent infinite tree. Each vertex is identified with a p-adic ball in $\mathbb{Q}_p$, and the tree encodes the p-adic valuation through branching depth. [established — mathematics; Kaletha & Prasad 2023]

The BT tree has the properties that Cartesian space lacks: - **No absolute origin:** The tree is a completely homogeneous graph — every vertex looks identical. Choosing an origin is an arbitrary observer choice, not a geometric fact. - **No inherent scale:** The tree's ultrametric distance is purely combinatorial — no "meter" or "second" is embedded in the tree itself. Scale emerges from the branching pattern, not from an external unit. - **Place-democratic:** The tree exists at every prime p. The Compton count's values at different primes are different projections of the same rational number. [speculative]

4.2 External Validation: p-adic AdS/CFT

The strongest external support for the BT tree as a physical geometry comes from the p-adic AdS/CFT program (2016-2021). Five papers in JHEP and Physical Review Letters independently developed the BT tree as a holographic tensor network [established — Bhattacharyya et al. 2018, Chen et al. 2021]:

- **Bhattacharyya, Hung, Lei, Li** [2018, JHEP, DOI 10.1007/jhep01(2018)139]: tensor network realisation of p-adic AdS/CFT on the BT tree
- **Hung, Li, Melby-Thompson** [2019, JHEP, DOI 10.1007/jhep04(2019)170]: proved the path integral of a p-adic CFT is equivalent to a tensor network on the BT tree
- **Chen, Liu, Hung** [2021, PRL, DOI 10.1103/physrevlett.127.221602]: emergent Einstein equations from p-adic CFT tensor networks

These papers use the BT tree for gravity (spacetime emergence from boundary CFT). ODR uses the same tree for particle ontology (mass identity as combinatorial coordinate). This is a consilience: two independent research programmes converge on the same mathematical structure from completely different physical questions. [speculative — application to particle ontology is novel]

4.3 Lorentz Invariance

The BT tree's ultrametric distance breaks continuous Lorentz symmetry — the tree is not isotropic. The thesis proposes that Lorentz invariance is an emergent Archimedean symmetry, recovered at large separation scales where the tree's discrete structure is smoothed into the real continuum. [speculative — no proof exists]

The p-adic AdS/CFT program faces the same issue: the BT tree's isometry group is $\mathbf{PGL}(2,\mathbb{Q}_p)$, not the Poincaré group. Yet the program successfully recovers gravitational dynamics. This precedent supports the claim that Lorentz breaking is UV-only and undetectable at accessible energies. [speculative]

Falsifiability: RQ3 would be disconfirmed if Lorentz violation were detected at energies below the Planck scale AND shown to have no tree-geometry structure — i.e., if the symmetry breaking were smooth and continuous rather than prime-structured. Existing bounds from Fermi-LAT (GRB photon arrival times) constrain any deviation to $\xi < 10^{-20}$ [established], consistent with tree-discreteness at Planck scale. This cannot DISPROVE the thesis — it is a necessary condition for the thesis to survive.

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5. RQ4: Testable Predictions

5.1 The Enabling Constraint

Without at least one falsifiable prediction that distinguishes ODR from standard QFT, the thesis is Philosophy rather than Physics. Five candidates have been enumerated (see `artifacts/rq4-candidate-predictions.md`). Two are prioritised here:

5.2 P2: Lorentz-Violation Dispersion from Tree Discreteness

The BT tree's ultrametric distance predicts a modified dispersion relation for particles:

$$E^2 = p^2c^2 + m^2c^4 + \xi \cdot \frac{E^3}{m_P}$$

where ξ is a dimensionless parameter related to the tree's branching factor p. [speculative — no calculation exists yet]

The prediction: gamma-ray burst (GRB) photons of different energies arrive at different times due to energy-dependent light speed. Fermi-LAT has detected approximately 10^4 GRBs with photon energies up to ~100 GeV. Current bounds constrain $\xi < 10^{-21}$. If the tree geometry produces a prime-dependent threshold structure — $\xi(2) \neq \xi(3)$ — the signal distinguishes tree-geometry from generic Lorentz invariance violation (LIV). [speculative — requires detection or stronger bounds]

5.3 P5: Quasiparticle Invariance Null Test

The quasiparticle extension predicts: no quasiparticle will ever be found with an invariant effective mass independent of host material. Systematic review of PDG and solid-state databases would test this negative prediction — it has held for approximately 90 years since Landau's Fermi liquid theory. A single counterexample (identical m^* in two chemically distinct lattices) would disconfirm the claim. [speculative]

Testability: P2 is testable within 2-5 years with existing Fermi-LAT data or the Cherenkov Telescope Array (CTA). P5 is testable now from PDG data — the absence of host-independent quasiparticles is the null result that holds. P1

(Compton-count quantisation in UHECR GZK cutoff), P3 (BT-tree scattering using Romanov-Rudin 1995 formalism), and P4 (Feynman diagrams as tensor network contractions) require further development.

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6. RQ5: The Silo Failure Diagnosis

6.1 The Five Trees

Five independent disciplines discovered the same $(p+1)$-valent combinatorial tree with cross-ratios between 1916 and 2016. Each called it by a different name. None connected it to particle ontology. The silo costs are staggering:

Domain	Structure Name	Year	Connection	Silo Cost	Key Pap
Number Theory	Ostrowski completions	1916	2026 (ODR)	**110 yr**	Ostrowsk Acta Mat 1916
Quantum Foundations	Zitterbewegung/Compton	1928	2026 (ODR)	**98 yr**	Dirac, Pr Roy. Soc 1928
Information Theory	Radix tree / trie	1960	2026 (ODR)	**66 yr**	Fredkin, 1960
Mathematical Physics	Bruhat–Tits tree	1980s	2016 (AdS/CFT)	~30 yr	Vladimir Volovich 1994
Holography	BT tensor network	2016	SAME ERA	~10 yr	Heydema et al., 20 2018

[established — historical; all dates and publication data verified]

The p-adic AdS/CFT program (2016-2021) partially closed the math-physics-to-holography gap (~30yr). But the particle-ontology connection — Compton counts to BT tree vertices — remains disconnected from every other domain. The thesis makes this connection explicit for the first time. [speculative]

6.2 Why Does This Matter?

Silo failures are not curiosities — they are productivity losses. The 110-year gap between Ostrowski's theorem (1916) and its application to particle ontology (2026) represents a century of physics done on $\mathbb{R}$ when the natural number system for particle identities was $\mathbb{Q}$ with its p-adic completions. The 66-year gap between the radix trie (1960, computer science) and BT-tree particle classification means six decades of particle classification without a natural combinatorial data structure. [established — the temporal gaps are historically verified; the interpretation of their significance is speculative]

The Consilience Gate (KIF-29, HARD) mandates that any QNFO publication spanning 2+ domains must produce a cross-domain lexicon, a minimum of one structural isomorphism per domain, and a silo cost table. This paper satisfies all three. The meta-principle that emerges: the Bruhat–Tits tree is the natural coordinate system for any physical quantity that admits a prime factorisation — Compton counts, entanglement entropies, particle masses, and coupling constants. The tree is scale-free, origin-free, and completion-agnostic. [speculative]

Falsifiability: RQ5 would be disconfirmed if a sixth domain were identified that uses the same tree AND was already connected to particle ontology before 2026. The literature search across 8 sources (OpenAlex, Crossref, Zenodo, Europe PMC, arXiv, QNFO Vectorize, QNFO KG, web search) found zero such connections. [established — Phase 1-2 due diligence of odr-thesis project]

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7. Synthesis and Conclusion

The ODR thesis proposes a coherent reformulation of particle identity: a particle IS its Compton count — a rational number whose prime factorisation assigns it a vertex on the Bruhat–Tits tree. The tree replaces $\mathbb{R}^3$ as the natural coordinate system. Quasiparticles are rational functions of background Compton counts, not invariant vertices. The CMB is a Gibbs distribution over photon Compton counts, not a fundamental entity. Five disciplines independently discovered the same tree without connecting it — ODR rectifies this century-long silo failure.

The thesis is weakest at RQ3 (BT tree as coordinate replacement — Lorentz recovery unproven) and RQ4 (prediction — candidates exist but none calculated yet). It is strongest at RQ1 (Compton ontology — self-consistent, quasiparticle extension handles condensed matter) and RQ5 (silo failure — slam-dunk historical diagnosis, zero external citations found at the intersection).

RQ4 candidate P2 (Lorentz-violation dispersion from tree discreteness, testable with Fermi-LAT GRB data within 2-5 years) is the fulcrum: if confirmed, it elevates RQ1-3 from Philosophy to Physics. If constrained below $\xi < 10^{-22}$ without a signal, the thesis survives (tree-discreteness at Planck scale is consistent with no signal at current precision), but its most testable prediction is eliminated, leaving only the null test (P5) and theory-development candidates (P3, P4).

The paper's claims are explicitly speculative where they exceed established physics, and all sections carry falsifiability conditions in accordance with qnfo-core §0.0. The Consilience Gate passes with a five-domain silo cost table, one structural isomorphism per domain, and a frontier question: can the Feynman diagram be reinterpreted as a tensor network contraction on the Bruhat–Tits tree?

Declarations

Author Contributions: QNFO: conceptualisation, formal analysis, writing.

Data Availability: No experimental data were generated. All cited values are from published sources validated against Crossref or DataCite metadata.

AI Assistance Disclosure: This paper was drafted with AI assistance under human direction.

Competing Interests: None declared.

Funding: None.

License: QNFO Unified License Agreement (QNFO-ULA).

Peer Review: This preprint has not undergone formal peer review.

Errata: No errata at time of publication. Corrections will be tracked in the project repository at github.com/QNFO/odr-thesis.

Version: v1.0, 2026-08-03.

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